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Studierichting: Informatica

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$$f : \Omega \subseteq \mathbb{R} \rightarrow \mathbb{R}^4 : x \rightarrow (2^{\sec x}, \coth x, \log_5 \sqrt{\cos x}, x^{x^x})$$

$$\frac{\partial}{\partial x}(2^{\sec x}) = 2^{\sec x} \cdot \ln 2 \cdot \sec x \tan x$$

$$\frac{\partial}{\partial x}(\coth x) = -\operatorname{csc}^2 x$$

$$\frac{\partial}{\partial x}(\log_5 \sqrt{\cos x}) = \frac{-\tan x}{2 \ln 5}$$

$$\frac{\partial}{\partial x}(x^{x^x}) = \frac{\partial}{\partial x}(e^{x^x \ln x})$$

$$= e^{x^x \ln x} \cdot \frac{\partial}{\partial x}(x^x \ln x)$$

$$= x^{x^x} \cdot \frac{\partial}{\partial x}(x^x \ln x)$$

$$= x^{x^x} \left(\frac{x^x}{x} + x^x \cdot (\ln x + 1) \cdot \ln x \right)$$

$$= x^{x^x} (x^x (\ln x + 1) \ln x + x^{x-1})$$

$$Df(x) = \begin{pmatrix} 2^{\sec x} \cdot \ln 2 \cdot \sec x \tan x \\ -\operatorname{csc}^2 x \\ -\frac{\tan x}{2 \ln 5} \\ x^{x^x} (x^x (\ln x + 1) \ln x + x^{x-1}) \end{pmatrix}$$

References